Unit 1

Lesson 3

Spark The Challenge!

Building Our Green Future.

Time Frame: [90 min].

Lesson Objectives:

By the end of this lesson students will be able to:

- Build a 3D smart bin model in Tinkercad, using the Ruler tool to measure precisely.
- Use perimeter and area formulas to figure out the correct dimensions for your bin.
- Understand how to identify wet and dry waste and sort each type correctly.

Challenge questions of the lesson

1. **How can we creatively design and improve a "smart bin" with additional, effective features to solve real-world waste management problems like pollution and energy shortages?**
2. **Why is precise measurement crucial when designing 3D models in Tinkercad, and how do tools like the Ruler ensure components fit and function correctly in a smart device?**
3. **What is the fundamental importance of separating wet and dry waste, and how does this process contribute to sustainable waste management and giving materials a "new life"?**
4. **How can our knowledge of technology (like sensors, LEDs, coding, and 3D modelling) be integrated with an understanding of environmental challenges to build practical, automated solutions?**

SEL Relation

- **Communication (Sharing our ideas):** We will practice sharing our awesome smart bin designs and "Super-Guides," explaining our creative choices and how they help tackle real-world waste problems. Time to show off our brilliant thinking!
- **Respect (Listening to others):** We will listen carefully and respectfully when our classmates present their fantastic work, cheering them on and learning from their ideas.
- **Empathy (Thinking about others' feelings)**: We will think about how our smart bin designs and "Super-Guides" can make it super easy and clear for everyone to sort trash correctly, helping them feel good about doing their part for our planet!
- **Self-Awareness (Understanding our own work):** We will reflect on all the cool design choices we made for our smart bin and "Super-Guide," understanding why those choices are super important for keeping our environment healthy and our waste sorted perfectly.

Engage

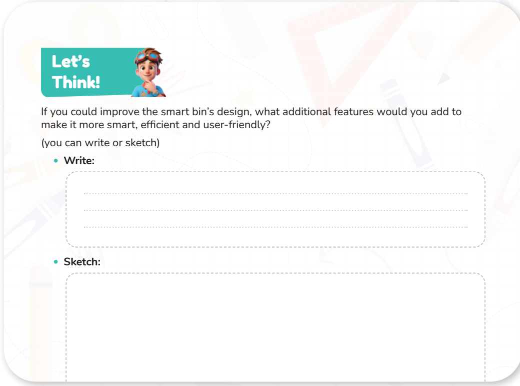
Teacher Role:

- **Storyteller:** Begin by reading or narrating a story that presents a real-world Story **Connector:** Begin the session by reading the short dialogue story aloud to re-engage students with the idea of collaboratively designing their "smart trash bin."
- **Activator of Prior Knowledge:** Ask guiding questions to connect to past experiences:
  1. “Do you remember learning about 3D models or shapes that could become real objects?”
  2. “Why do you think separating wet and dry waste is that important for our smart bin?”
- **Facilitator:** Explain the bin design activity clearly and guide access to design tools (digital). Guide students to think creatively about how shapes and features can represent parts of a functional smart bin.
- **Encourager of Design Thinking:** Prompt students to think like young engineers solving a real problem:
  1. “What can these shapes become on our smart bin? Could this be a new type of lid, a special section for paper, or where a sensor might go?”
  2. "How can we make sure our bin sorts trash even better?"
- **Supporter of Expression:** Allow multiple ways for students to express their ideas through drawing, labeling, or explaining their designs to peers.

Expected From Students:

- **Active Listening:** Listen to the story and imagine being part of the smart bin design team with Mrs. Sara, Adam, and Laila.
- **Creative Thinking:** Think about what different shapes and design elements remind them of and how they could be part of a functional and user-friendly smart trash bin.
- **Hands-On Making:** Sketch or draw a design for a smart trash bin, carefully incorporating sections, lids, or other features they imagine. (Referencing for visual guidance on drawing bin designs).
- **Application of Learning:** Use prior knowledge of waste sorting and bin functionality to make their smart bin design effective for managing trash (clear sections for wet/dry, easy opening lids).
- **Visual Communication**: Color and label their smart bin design clearly so others can understand what they made and how it helps sort trash.

Parts of the books included.



Story

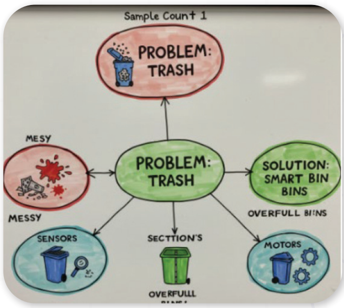
1. **Set the Scene with Excitement and Purpose.**
   - Teacher says (enthusiastically): "Mrs. Sara's class is back today—and they're ready to solve a real-life mystery in their schoolyard! Can you guess what problem they're going to tackle, and what amazing solution they might design?"
   - Ask students: "If you could design a 'smart bin' or a better system to keep our schoolyard super clean, what ideas would you include?"
     - (Let students respond, activating their imagination and connecting personally to the topic of waste solutions.)
2. **Read the Story Aloud with Voice and Energy.**
   - Use tone, facial expressions, and character voices to bring the story to life:
     - Use a thoughtful voice for Mrs. Sara.
     - Use worried, then curious voices for Adam and Laila.
3. **Ask a Think-Aloud Question to Prompt Engagement.**
   - Teacher says: "Wow! Laila and Adam are thinking like real engineers about making a 'smart bin.' What do you think they should add to their bin or their system to make sure it handles all the trash properly and easily, so our schoolyard can be clean?"
   - Encourage student responses like "clear labels," "different sections for recycling," "easy-to-open lids," or "a way to tell us when it's full."

4. **Transition to the Design Activity.**
   - Teacher says: "You're going to be like Mrs. Sara's class today! Let's start designing our own ideas for a better waste system. You'll get to create something that could make our school yard much cleaner!"
   - Show the paper with space for their bin designs [Let's think part] and explain the activity steps briefly before passing them out.

Possible Strategies for Implementation:

Possible Strategies for Implementing the Story

1. **"Problem Walkthrough"**
   - Visually demonstrate or physically represent the "messy schoolyard" scenario using props or a simple drawing on the board before reading the story. This helps make the problem tangible and immediately relatable for students.
2. **"Emotion Check-In"**
   - Pause after describing the messy schoolyard in the story and ask students: "How do you think Adam and Laila felt when they saw the messy yard? How would you feel?" This builds empathy and connects them to the problem emotionally.
3. **"Keyword Web"**
   - As you read the story, jot down keywords or simple drawings on a whiteboard or chart paper to create a "problem web" or "solution idea web." For example, "Trash," "Messy," "Overfull Bins," leading to "Smart Bin," "Sections," "Labels."

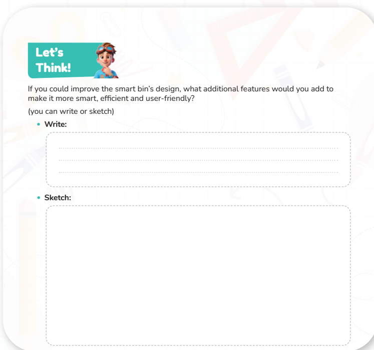


4. **"Quick Idea Share"**
   - After Mrs. Sara introduces the idea of a "better system" or "smart bin," have students quickly turn and tell a partner "One smart idea" they have for the bin before the teacher asks the full Think-Aloud Question. This allows for immediate, low-stakes participation.

Let's Think!

Objective:

Students will creatively apply their understanding of waste management and design thinking by proposing additional features for a "smart trash bin," focusing on making it more efficient, user-friendly, and intelligent. Through sketching and/or writing, they will refine their problem-solving skills and connect concepts to practical applications.



Possible Answers:

→ Write:

I would add a sound alert to let people know when the bin is full.

Also, I would include a foot pedal in case the sensor stops working.

The bin could have a screen that shows what type of waste goes inside.

We can even add a solar panel to save energy!

→ Sketch:

(For the sketch area, students might draw a rectangular bin with two compartments labeled "Wet" and "Dry," a small solar panel on top, an LCD screen in front, and sound waves next to a speaker. A tiny foot pedal could also be shown at the bottom front.)

Activity Instructions:

1. **Introduce the Activity.**
   - Introduce the Activity: Let's Make It Smarter!
     - Say: "Today, we're going to take our 'smart bin' design to the next level! We've already thought about sorting trash, but now, 'Let's Think!' about how we can make our smart bin even more amazing, efficient, and super easy for everyone to use!"
     - Show the "Let's Think!" prompt and review the question: "If you could improve the smart bin's design, what additional features would you add to make it more smart, efficient and user-friendly?"
2. **Distribute Materials**
   - Give each student:
     - Ask them to open the task from their books.
     - Pencils or crayons/markers.
3. **Explore and Imagine: Feature Brainstorm!**
   - Prompt students:
     - "Think about the challenges of trash in our schoolyard. What frustrates Mr. Yusef? What makes it hard for students to sort?"
     - "How could a really smart bin help with those problems? Could it talk? Could it light up? Could it tell us something?"
     - "Imagine you're designing for the future! What magical or high-tech features would you add?"
   - Let students ponder ideas for a few minutes.
4. **Design and Detail: Write and Sketch Your Ideas!**
   - Have students use the provided space to:
     - Write: Describe their additional features. Encourage them to explain what the feature does and why it makes the bin "smarter," "more efficient," or "user-friendly."
     - Sketch: Draw their new features on the bin or show how they would work.

- Encourage them to give their new features names or describe what they do.

5. **Share and Reflect: Our Innovative Solutions!**
   - Invite students to:
     - Pair-share with a classmate, describing their proposed improvements.
     - Say how their idea helps solve a problem or makes the smart bin better.
     - "What makes your idea 'smart,' 'efficient,' or 'user-friendly'?"

Possible Strategies for implementation:

1. **"Feature Focus Chart" (Structured Brainstorming):**
   - Create a simple anchor chart with columns like: "Problem My Feature Solves," "My New Feature Idea," "How It Works," "Why It's Smart/Efficient/User-Friendly." This helps organize their thinking.
2. **"Role-Play a Problem" (Empathy-Driven Design):**
   - Briefly act out a common trash-related "problem" in the classroom (Try to put a juice box in the paper bin or a tissue, a bin overflowing). Then, ask: "How would your improved smart bin help in this situation?" This grounds their design in real challenges.
3. **"Future Tech Talk" (Inspiring Innovation):**
   - Briefly discuss examples of "smart" technology they might already know (a smart light that turns on when you walk by, a robot vacuum). Connect this to how a smart bin could use similar ideas to "think" or "help."
4. **"Design Gallery Walk" (Peer Inspiration):**
   - After students have sketched/written some ideas, have them quietly walk around and view their classmates' "Let's Think!" sheets. This can inspire new ideas or different ways to approach their own design before finalization. (make sure that they will not copy).

Explorer's Toolkit

Teacher Role:

- **Introducer:** Let's look at this picture. It shows a design for a recycling bin. We often just think about throwing things away, but this design makes us consider how we throw things away. It's about being smart with our waste.
- **Facilitator:** Guide students to understand what this bin is trying to do. How does separating "wet" and "dry" waste help? What makes it "smart"? Help them see how small changes in design can make a big difference for our environment.
- **Encourager:** Encourage students to think beyond the obvious. What other "smart" features could we add to a bin to make recycling easier or more effective? Why would those features be good?

- **Clarifier:** Make sure students understand the difference between "wet" waste (like food scraps) and "dry" waste (like paper or plastic). Why is it important to keep them separate?
- **Sharer:** Create a space for students to share their ideas for "smart" bin features. Ask them to explain why their ideas are helpful for recycling.

Expected From Students:

- **Understanding the Concept:** Students will get that sorting waste, especially "wet" and "dry," is important for recycling. They'll also see that design and technology can help with this.
- **Feature Association:** Students will think about what "smart" features could make a recycling bin work better for people and the planet.
- **Creative Decision-Making:** Students will come up with their own unique ideas for making a recycling bin "smart" or more useful.
- **Hands-on Participation:** Students will actively brainstorm, discuss, and suggest ways to improve waste sorting and recycling through design.
- **Articulation of Reasoning:** Students will be able to explain why they chose specific "smart" features and how those features would help with recycling or environmental efforts.

Parts of the books included.


Explorer's Toolkit

DIY Lab

Let's dive into Tinkercad at https://www.tinkercad.com and start designing a trash bin that's more than just a bin—it's smart and built to help the planet!

We'll create each part with features that make sorting and recycling easier—and even a little fun!

Figure 10: Recycling Bin Model Created in Tinkercad.

DIY Lab

Objective:

Students will explore how thoughtful design choices, especially with technology, can create smarter solutions for everyday problems like waste management. By designing components and imagining smart features for a recycling bin, students will practice critical thinking, problem-solving, and practical design, all while learning how innovation can make a real difference for our environment.

Activity Instructions

Step 1: Set the Context.

- Say: "Today, we're going to be design thinkers! We've all seen trash bins, but what if we could make them smart? Look at this picture of a smart bin. It separates 'wet' and 'dry' waste. Good design can really help us recycle better and keep our planet cleaner."

Step 2: Explain the Task.

- Read the directions for the activity.
    - Say: "You're going to plan your own Smart Bin Design. You'll think about the main parts, and then brainstorm cool features that make recycling easier. Instead of coloring, you'll be writing down your ideas and maybe sketching them out."

Step 3: Materials.

- Laptops
- Internet connection
- Access to Tinkercad (instructions to open it).

Step 4: Work Time.

Students will apply the design on tinkercad:

Software link:https://www.tinkercad.com/

Activity Instructions

Access Tinkercad: Open the Tinkercad website or app and log in (students need class code and username should be pre-prepared).

- Objective: Students will explore how thoughtful design choices, especially with technology, can create smarter solutions for everyday problems like waste management. By designing components and imagining smart features for a recycling bin, students will practice critical thinking, problem-solving, and practical design, all while learning how innovation can make a real difference for our environment.

Activity Instructions.

Step 1: Set the Context

- Say: "Today, we're going to be design thinkers! We've all seen trash bins, but what if we could make them smart? Look at this picture of a smart bin. It separates 'wet' and 'dry' waste. Good design can really help us recycle better and keep our planet cleaner."

Step 2: Explain the Task.

- Read the directions for the activity.
- Say: "You're going to plan your own Smart Bin Design. You'll think about the main parts, and then brainstorm cool features that make recycling easier. Instead of coloring, you'll be writing down your ideas and maybe sketching them out."

Step 3: Materials.

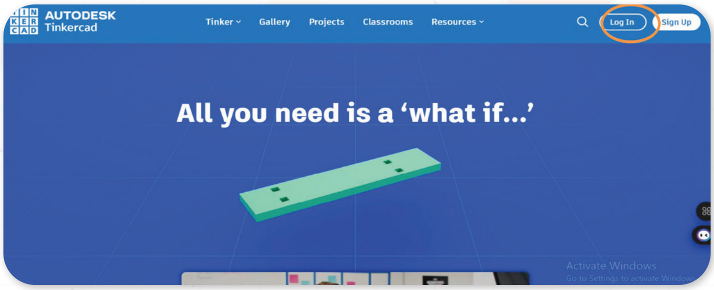
- Laptops
- Internet connection
- Access to Tinkercad (instructions to open it).

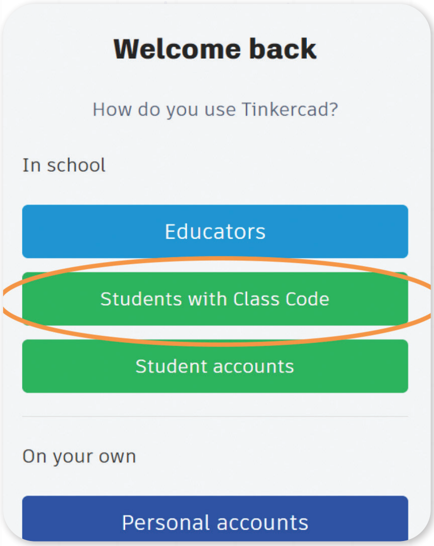
Step 4: Work Time.

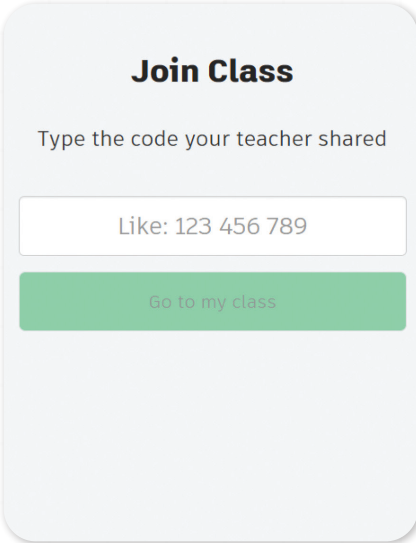
- Students will apply the design on tinkercad:
- **Software link:** https://www.tinkercad.com/

Tinkercad Instructions

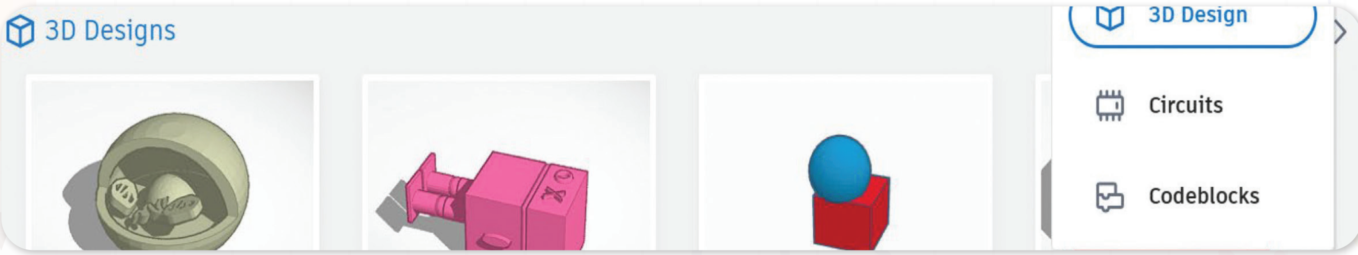
1. Open the Tinkercad website or app and log in (students need class code and username should be pre-prepared).



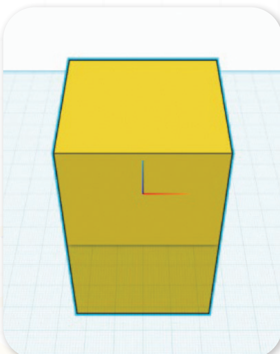




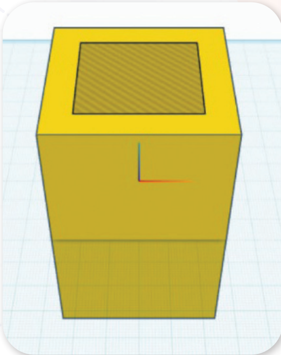
2. Create New Design: Start a new 3D design.



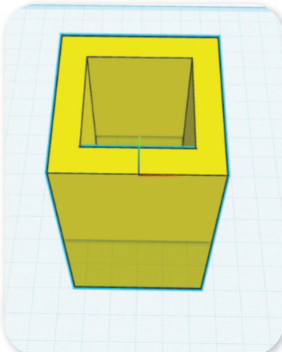
A. The first step: drag and drop a cube (60*60*100) (w*l*h) then change its colour to yellow (you can change the size and colours).



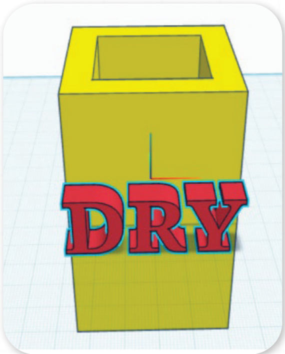
B. The second step: drag and drop another box(hole) and put it inside the main box (40*40*90).



C. The third step : make them as a group.



D. The fourth step: drag and drop a text and change it into “DRY”



Possible Strategies for implementation:

1. Teacher Modeling:

- Fill in one example on the screen:
  - "I'm going to start by making the 'dry' trash, then what do think the next step will be?"
- Show a basic Tinkercad step (like creating the first box).
- You can complete the design in the same way or flip it to the students but make sure to mentor them

2. Flexible Design Options:

- Allow students to:
  - Use Tinkercad to create their main bin.
  - Sketch out complex "smart" features on paper first if that helps their thinking.
  - Add notes or labels within Tinkercad if they can.

3. **Reflection and Sharing:**
   - After designing, students can:
     - Show their Tinkercad designs to the class or a partner.
     - Explain the "smart" features they included and why those features are important for better recycling.

Explain

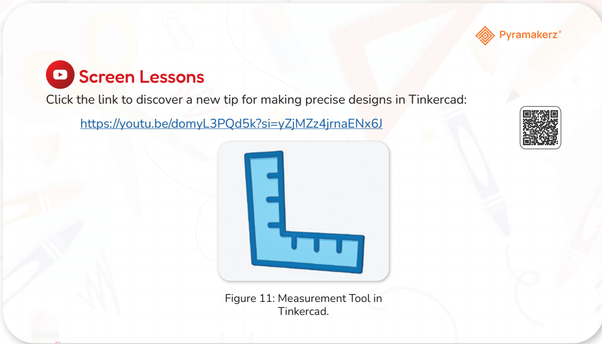
Teacher Role:

- **Introducer:** "Today, we're going to dive into Tinkercad and learn a super important tool for precise design: the Ruler! This video will show us exactly how to use it the easy way, which is key for building accurate models like our smart recycling bins."
- **Facilitator:** Guide students through the video, pausing to highlight how the ruler helps engineers and designers get exact measurements and align objects perfectly. Emphasize how this precision helps catch problems early in the design phase.
- **Purpose Setter:** Explain that the video demonstrates how to use the Tinkercad Ruler tool. "Learning this tool will be essential for making sure the different parts of our smart recycling bin fit together perfectly, like the separate compartments for wet and dry waste."
- **Engagement Encourager:** Prompt students to think about why having exact measurements is so important when designing real-world objects, and how the ruler makes this easier.
- **Discussion Moderator:** Lead a discussion about the different modes of the ruler (endpoint vs. midpoint) and when each is most useful.
- **Concept Clarifier:** Answer any questions about the ruler tool, its functions, and how it applies to their specific smart recycling bin design.
- **Connection Maker:** Help students connect the video's ruler techniques to their smart recycling bin design in Tinkercad. "How will using the ruler help us design a better, more functional bin?"

Expected From Students:

- **Active Viewing:** Students will watch the video carefully, focusing on the steps and tips for using the Tinkercad Ruler tool.
- **Observation:** Students will identify the different functionalities of the ruler (placing, moving origin, endpoint/midpoint modes, precise input).
- **Thoughtful Participation:** Students will share their ideas about how the ruler tool can be used to improve the accuracy and design of their smart recycling bin project.
- **Critical Analysis:** Students will think critically about how precise measurement, enabled by the ruler, helps create better solutions for real-world design problems

Parts of the books included.



Screen Lessons

Objectives:

Students will understand the fundamental concepts and tools used in 3D modeling, specifically focusing on the precise use of the Ruler tool in Tinkercad, preparing them to apply these digital design skills to create their smart recycling bin.

Video link: https://youtu.be/domyL3PQd5k?si=yZjMZz4jrnaENx6J

Possible Strategies for implementation:

5. **Pre-Watching: Set a Purpose**

Strategy: What Makes a Good Fit?

- How to Implement:
  - On the board, draw two columns: "Things That Need to Fit Perfectly" and "Why Precision Matters."
  - Ask: "Think about things in the world that need to fit together just right (e.g., a lid on a box, a door in a frame). What happens if they don't fit?" (Add examples to the first column: phone cases, LEGOs, puzzle pieces, our smart recycling bin's compartments).
  - Then ask: "Why do you think it's important for designers to be super precise with measurements?" (Add ideas to the second column: parts won't work, waste materials, looks messy).
  - Tell students: "We're going to watch a video that shows us the most important tool in Tinkercad for making things fit perfectly: the Ruler!"
- Benefit: Activates their thinking about precision in design, introduces the practical need for a tool like the ruler, and sets a clear reason for watching the video.

6. **During Watching: Active Engagement.**

Strategy: Ruler Feature Checklist

- How to Implement:
  - Give each student a simple checklist or let them use whiteboards with sections for: "How to place the ruler," "How to move an object to the origin," "What are midpoint/endpoint modes?", "How to type in exact sizes."

  - Ask them to watch closely and tick off or write a quick note as each feature of the ruler is demonstrated and explained in the video.
  - Challenge them to think about how each feature could be used in their smart recycling bin design (e.g., "I could use endpoint mode to make sure the bin's base is exactly 50mm wide," or "Midpoint mode will help me centre the opening for trash").
- **Benefit:** Encourages active listening for specific instructions and functionalities, reinforces the video's key steps, and directly connects the tool's features to their design task.

7. **Post-Watching: Discussion and Reflection.**

Strategy: "My Ruler Plan" Sentence Frame.

- How to Implement:
  - Ask students to complete and share this sentence: "I learned that the Tinkercad Ruler helps me [mention a specific ruler function from the video, 'align objects perfectly' or 'measure from the centre'], so I can use it to design my smart recycling bin by [explain how they will apply it,'making sure the sensor opening is exactly centred' or 'creating compartments of specific dimensions']."
  - Example: "I learned that the Tinkercad Ruler helps me use midpoint mode, so I can use it to design my smart recycling bin by placing the circular opening for the trash exactly in the middle of the lid."
- **Benefit:** Reinforces their understanding of the ruler's practical uses, encourages them to link the video's theory to their project, and builds confidence in applying new knowledge.

Dig Deeper

Objective:

Students will develop a deeper understanding of the critical importance of waste segregation (dry and wet waste) and its role in sustainable waste management. By actively engaging with an article through anticipation, focused observation, and structured reflection, they will practice critical thinking, listening, and communication skills as they analyse how proper sorting gives materials a "new life," revise initial assumptions, and articulate new insights about environmental responsibility.

Article link: https://www.eletimes.com/analog-vs-digital-electronics-difference-and-comparison

Possible Strategies for implementation.

1. **Pre-Reading: Activating Prior Knowledge & Setting Purpose**

Strategy: KWL Chart - "Recycling Center Expedition"

- How to Implement:
  - Introduce the article topic: "Today, we're going to read an article all about how we separate our trash into 'wet' and 'dry' waste. This is super important for recycling and managing our trash. Before we start, let's think about what we already know and what we want to find out."
  - “K" (Know): Ask students, "What do you already know about separating trash, like 'wet' and 'dry' waste, or why we even do it?" Have them individually fill in the "K" column with their existing knowledge.
  - “W" (Want to Know): Next, ask, "What do you want to learn or wonder about separating wet and dry waste, or how it helps?" Have them fill in the "W" column.
  - Explain that reading the article will help them fill in the "L" column later.

Please: print them the following chart:

2. During Reading: Focused Information Gathering.

Strategy: Guided Reading with Highlighting/Note-Taking - "Waste Segregation Detectives"

- How to Implement:
  1. Instruct students to read the article "Dry Waste and Wet Waste Management | Waste Segregation - Shakti Plastic Industries" either individually, in pairs, or as a class.
  2. Provide specific prompts to guide their reading. Examples:
  - "As you read, try to find out the main differences between 'wet' and 'dry' waste."
  - "Look for why separating these types of waste is so important."
  - "Pay attention to how waste segregation helps the environment or creates new resources."
  3. Encourage students to highlight or jot down brief notes/keywords related to these prompts directly on a paper or in their digital document.

3. **Post-Reading: Consolidating Learning & Reflection.**

Strategy: KWL Chart Completion & Discussion - "Waste Experts Share"

- How to Implement:
    1. Bring students back together.
    2. "L" (Learned): Ask students to fill in the "L" column of their KWL chart with new information they learned from the article, paying special attention to answering questions from their "W" column.
    3. Share and Discuss: Facilitate a whole-class discussion where students share what they wrote in their "L" column.
    4. Reflect: Ask concluding questions: "What was the most surprising thing you learned about separating wet and dry waste?" or "How does knowing more about waste segregation.

Elaborate and Evaluate

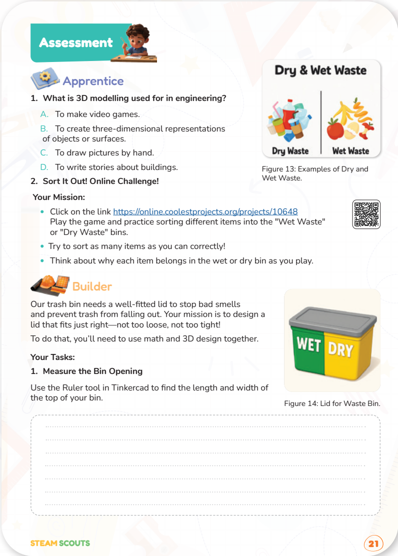
Teacher Role:

- **Clarifier:** Ensure students understand the difference between 2D (flat) shapes and 3D (solid) shapes, revisiting prior lessons or the video as needed.
- **Facilitator:** Guide students in using Tinkercad or other digital tools to explore and build with 3D shapes.
- **Encourager:** Support creative thinking by reminding students of the goal—designing a chair for families that is both fun and functional.
- **Supporter:** Help students navigate basic functions in Tinkercad (e.g., resizing, rotating, combining shapes).
- **Observer:** Watch how students identify shapes, choose shapes that roll or stack, and apply this knowledge while designing. Take note of those who show understanding and those who need further support.

Expected From Students:

- **Shape Identification:** Students will correctly identify examples of 2D and 3D shapes and describe their features (e.g., "a circle is flat," "a sphere can roll").
- **Shape Application:** Students will predict which 3D shapes can roll, stack, or support weight—connecting this knowledge to design thinking.
- **Digital Modelling:** Students will use Tinkercad to create a simple 3D model of a small chair using basic shapes like cylinders, cubes, and rectangles.
- **Creative Design Thinking:** Students will show their understanding of shape properties by applying them purposefully in a model meant for real-world use (seating).

Parts of the books included.


Assessment

Apprentice

1. What is 3D modelling used for in engineering?
A. To make video games.
B. To create three-dimensional representations of objects or surfaces.
C. To draw pictures by hand.
D. To write stories about buildings.
2. Sort It Out! Online Challenge!

Your Mission:
- Click on the link https://online.coolestprojects.org/projects/10648 Play the game and practice sorting different items into the "Wet Waste" or "Dry Waste" bins.
- Try to sort as many items as you can correctly!
- Think about why each item belongs in the wet or dry bin as you play.

Dry & Wet Waste

Figure 13: Examples of Dry and Wet Waste.

Builder

Our trash bin needs a well-fitted lid to stop bad smells and prevent trash from falling out. Your mission is to design a lid that fits just right—not too loose, not too tight!

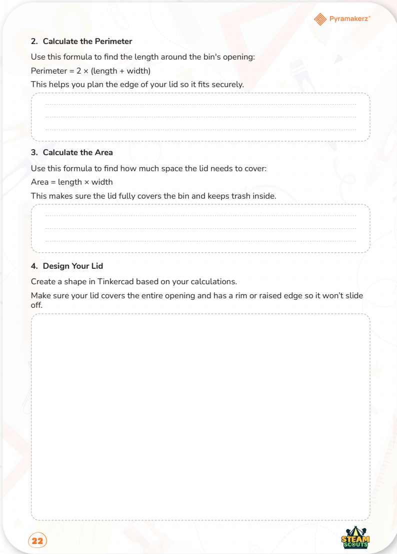
To do that, you'll need to use math and 3D design together.

Your Tasks:
1. Measure the Bin Opening

Use the Ruler tool in Tinkercad to find the length and width of the top of your bin.

Figure 14: Lid for Waste Bin.

STEAM SCOUTS 21


2. Calculate the Perimeter

Use this formula to find the length around the bin's opening:

Perimeter = 2 × (length + width)

This helps you plan the edge of your lid so it fits securely.

3. Calculate the Area

Use this formula to find how much space the lid needs to cover:

Area = length × width

This makes sure the lid fully covers the bin and keeps trash inside.

4. Design Your Lid

Create a shape in Tinkercad based on your calculations.

Make sure your lid covers the entire opening and has a rim or raised edge so it won't slide off.

22


Hey friends let's wrap up!

- Perimeter is the total distance around the edge of a shape.
- Area is the amount of space inside the shape.
- We need to learn about perimeter and area because they help us design things that fit just right—like making sure a trash bin lid covers the top properly or that parts of a project are the correct size. Whether we're building something in Tinkercad or solving real-world problems, knowing how to measure and calculate helps us plan, design, and build accurately.

Mastermind

You're becoming amazing at sorting trash. But what about others who don't know the difference between wet and dry? That's where you come in!

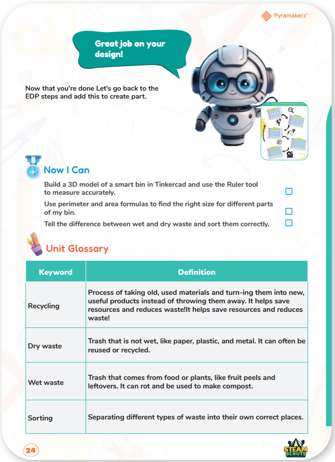
Your Mission:

Create a Super-Guide (like a mini-poster or logo) that teaches everyone how to sort their trash perfectly!
- Show what goes into the Wet Waste bin.
- Show what goes into the Dry Waste bin.
- Make it super clear and easy for anyone to understand.

Dry Waste

Figure 15: Wet and Dry Waste Examples.

STEAM SCOUTS 23


Great job on your design!

Now that you're done Let's go back to the EDP steps and add this to create part.

Now I Can
- Build a 3D model of a smart bin in Tinkercad and use the Ruler tool to measure accurately.
- Use perimeter and area formulas to find the right size for different parts of my bin.
- Tell the difference between wet and dry waste and sort them correctly.

Unit Glossary

Keyword	Definition
Recycling	Process of taking old, used materials and turn-ing them into new, useful products instead of throwing them away. It helps save resources and reduces wasteIt helps save resources and reduces waste!
Dry waste	Trash that is not wet, like paper, plastic, and metal. It can often be reused or recycled.
Wet waste	Trash that comes from food or plants, like fruit peels and leftovers. It can rot and be used to make compost.
Sorting	Separating different types of waste into their own correct places.

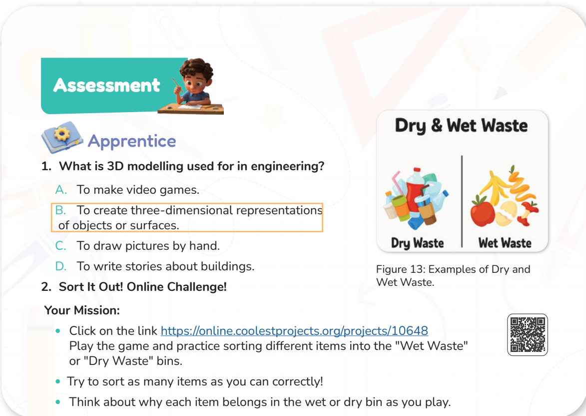
24

Apprentice

Objective:

Students will demonstrate their understanding of 3D modeling's purpose in engineering and apply their knowledge of dry and wet waste segregation by correctly sorting items in an online challenge, reflecting on their choices.

Answers:


Assessment

Apprentice

1. What is 3D modelling used for in engineering?
A. To make video games.
B. To create three-dimensional representations of objects or surfaces.
C. To draw pictures by hand.
D. To write stories about buildings.
2. Sort It Out! Online Challenge!

Your Mission:
- Click on the link https://online.coolestprojects.org/projects/10648 Play the game and practice sorting different items into the "Wet Waste" or "Dry Waste" bins.
- Try to sort as many items as you can correctly!
- Think about why each item belongs in the wet or dry bin as you play.

Figure 13: Examples of Dry and Wet Waste.

Possible Strategies for implementation:

For Task 1: "What is 3D modelling used for in engineering?"

1. **Quick Recall & Connect:**
   - How to Implement:
     - Before students answer, briefly revisit the core idea from our earlier video: "Remember when we watched the video about 3D modeling? What was one big reason engineers use it?"
     - Then, introduce the multiple-choice question.
   - Why it Helps: Reminds students of prior learning and helps them connect it directly to the question.
2. **Discussion & Justification:**
   - How to Implement: After students choose an answer, ask them to explain why they picked that option. Also, ask them to explain why the other options aren't quite right.
   - Why it Helps: Encourages deeper thinking, reinforces understanding, and helps clarify any misconceptions.

For Task 2: "Sort It Out! Online Challenge!"

1. **Game Introduction & Mission Briefing:**
   - How to Implement:
     - Introduce the "Sort It Out! Online Challenge!" game by sharing the link.
     - Read "Your Mission" together, emphasizing the part about "Think about why each item belongs in the wet or dry bin as you play."
     - Say: "This isn't just about speed; it's about making smart choices based on what we learned about wet and dry waste."
   - Why it Helps: Sets a clear purpose for the game beyond just playing, focusing on the learning objective.
2. **Play & Analyze:**
   - How to Implement:
     - Have students play the game individually or in small groups.
     - Encourage them to pause if they're unsure about an item and discuss it with a partner or jot down notes about tricky items.
   - Why it Helps: Provides hands-on practice, immediate feedback, and encourages problem-solving during the activity.
3. **Post-Game Reflection & Real-World Link:**
   - How to Implement:
     - After playing, bring students back together for a discussion.
     - Ask: "What items were tricky to sort? Why?"
     - Discuss common mistakes and correct them as a group, referring back to the "Dry Waste and Wet Waste" article if needed.

    - Finally, ask: "How does sorting in this game help us understand why our smart trash bin has 'wet' and 'dry' compartments?"
- Why it Helps: Consolidates learning from the game, addresses challenges, reinforces the importance of accurate sorting, and directly links the game to their Tinkercad project.

Builder

Objective:

Students will apply their understanding of 3D shapes and spatial reasoning to design a digital lid for their smart trash bin using Tinkercad. This activity will help them explore how shapes can be combined and manipulated to create functional components for their larger design. The goal is a lid that keeps yucky smells in and messy spills out, encouraging practical design thinking, creativity, and digital modeling skills.

Software link:

https://www.tinkercad.com/

Activity Instructions.

- Access Tinkercad: Open the Tinkercad website or app and log in (students need class code and username should be pre-prepared).
- Continue on the same design of the trash bins.

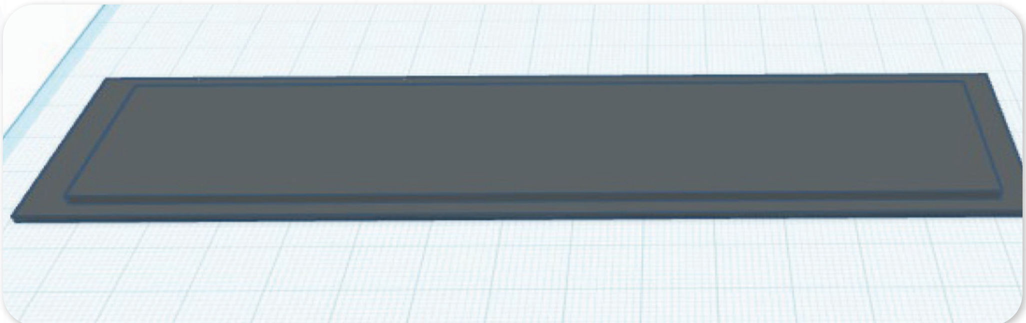
Your Tasks for Designing the Lid:

1. **Measure the Bin Opening:**
    - Use the Ruler tool in Tinkercad to find the length and width of the top of your bin.
    - Students should write down these measurements in their notebooks or on a worksheet).
    - (Guidance for AL/OL/BL can be integrated here).

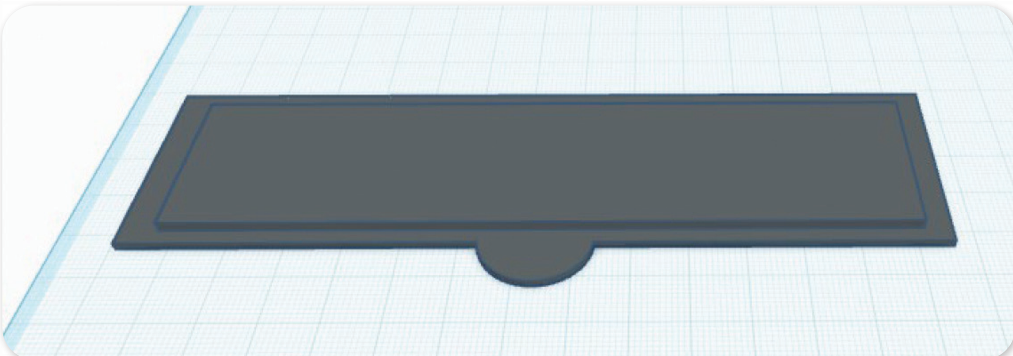
(AL might be given pre-measured dimensions, OL measures with guidance, BL measures independently).

2. **Calculate the Perimeter:**
    - Use this formula to find the length around the bin's opening:
        1. Perimeter = 2 × (length + width).
    - This helps you plan the edge of your lid so it fits securely.
    - (Students should write down their calculation).
3. **Calculate the Area:**
    - Use this formula to find how much space the lid needs to cover:
        1. Area = length × width.
    - This makes sure the lid fully covers the bin and keeps trash inside.
    - (Students should write down their calculation).

4. **Design Your Lid in Tinkercad:**
    - Create a shape in Tinkercad based on your calculations from steps 1-3.
    - The Top Piece: Start with a flat block (a Box shape). Adjust its length and width to match the measurements you took in Step 1 (or slightly larger, as specified by the teacher). Ensure it fully covers the bin's opening.
    - Add a Rim or Raised Edge: Make sure your lid has a rim or raised edge so it won't slide off the bin. This might involve adding another thin shape underneath the main lid piece or using the "Hole" feature to create an inset.
    - (Differentiated Instruction for this step:).
        1. AL (Approaching Level): Focus on making a simple flat top piece that covers the opening and adding a basic rim.



        2. OL (On Level): Design a simple lid that covers the opening and adds a small handle (a small cylinder, half-sphere, or other simple shape) and a functional rim or edge.



        3. BL (Beyond Level): Challenge students to try making the lid edges curved, or design a more complex rim structure that snaps into place or has a unique locking mechanism.

- Group and Personalize Your Lid:
    - Once all pieces of your lid are complete, select them all and group them together.
    - You can also paint your lid to match your bin or give it a unique colour. Think about what will make your lid look good and work best.

Reflection Questions:

- What shapes did you use to build your lid?
- Why did you choose those shapes for your lid?

- How does your 3D lid design help keep smells in and spills out for the smart bin?
- How is designing a 3D lid different from just drawing a lid on paper?

Mastermind

Objective:

Students will apply their understanding of wet and dry waste segregation by designing a clear, compelling "Super-Guide" (mini-poster or logo). This activity will help them practice effective visual communication, synthesize information, and encourage others to sort trash perfectly, reinforcing their role as environmental advocates.

Materials needed:

Paper, markers, colored pencils, or digital design tools (like Google Slides, Canva, paint, or a simple drawing app).

Activity Instructions:

1. Introduce the Mission: "You've become amazing at sorting trash. But what about everyone else who doesn't know the difference between wet and dry? That's where you come in! Your mission is to create a 'Super-Guide' – like a mini-poster or a logo – that teaches everyone how to sort their trash perfectly!"
2. Clarify Content: Remind students they need to:

- Show clearly what goes into the Wet Waste bin.
- Show clearly what goes into the Dry Waste bin.

3. Emphasize Clarity: "Make it super clear and easy for anyone to understand, even someone who has never sorted trash before!"

Possible Strategies for Implementation:

1. Content Brainstorm – "What Goes Where?"

- How to Implement:
  - o Start with a quick class discussion or have students jot down a list of common items.
  - o Ask: "What are some typical 'wet' waste items we see every day?" (food scraps, fruit peels).
  - o Then: "What about 'dry' waste?" (paper, plastic bottles, cardboard).
  - o This ensures they have plenty of examples for their guides.
- Why it Helps: Activates their knowledge about waste types, giving them the core content for their design.

2. Visual Inspiration Hunt – "Best Signs Ever!"

- How to Implement:
  - o Show examples of simple, effective signs, logos, or public service posters (not necessarily about recycling).

  - Ask: "What makes this sign easy to understand? What colors or symbols grab your attention?"
- Why it Helps: Gives students ideas for visual communication, helping them think about clarity and impact.

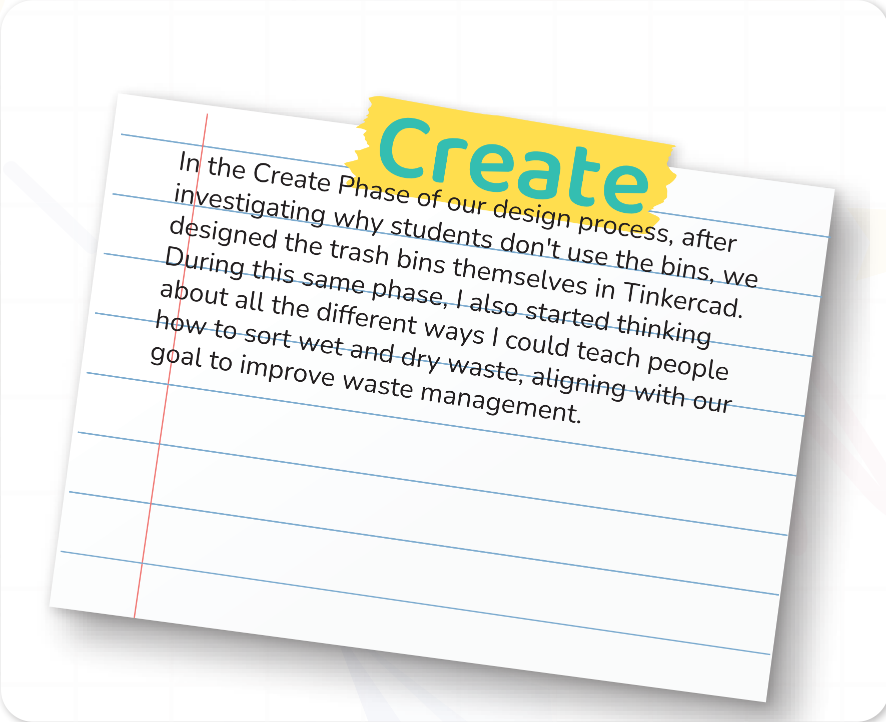
3. **Sketching & Planning – "Rough Draft First!"**
- How to Implement:
  - Before jumping into the final guide, have students do a quick rough sketch on scrap paper.
  - Encourage them to think: "Where will the 'Wet' bin info go? What image will I use for 'Dry'?"
  - Why it Helps: Allows for quick experimentation with layouts and ideas without worrying about perfection.

4. **Audience Check – "Would My Neighbor Get This?"**
- How to Implement:
  - As students work, remind them to think about their audience.
  - Ask: "If someone who knows nothing about sorting saw this, would they understand immediately?" "Is it clear for everyone?"
- Why it Helps: Encourages empathy in design and pushes for ultimate clarity in their message.

5. **Peer Feedback – "Fresh Eyes Help!"**
- How to Implement:
  - Once guides are somewhat complete, have students briefly show their design to a partner.
  - Partners can give constructive feedback, focusing on clarity: "I understood what goes in 'wet,' but maybe a different picture for 'dry' would make it even clearer."
- Why it Helps: Offers valuable insights from another perspective and helps students refine their communication.

6. **Presentation & Explanation – "Our Sorting Solutions!"**
- How to Implement:
  - Have each student or group briefly present their "Super-Guide" to the class.
  - Ask them to explain: "What are the most important things you want people to learn from your guide?"
- Why it Helps: Reinforces their learning, practices public speaking, and celebrates their creative solutions.

Don't forget to go back to the EDP page and add these to creating part.



Now I Can

Objective:

Students will reflect on their learning journey by identifying key skills used throughout our smart design project. This includes their understanding of how 3D design helps solve problems, and their ability to identify and differentiate between wet and dry waste. This activity promotes thinking about their own learning, helps them take ownership of their knowledge, and reinforces design, technology, and environmental understanding.

Activity Instructions:

1. Review Each Statement: Read through the checklist one sentence at a time. If you're not sure what something means, feel free to ask your teacher.
2. Check What You Can Do: Put a check mark in the box next to the things you feel confident about or remember understanding and doing during our project.
3. Think Back to Your Work: Look at your Tinkercad smart bin design, your "Super-Guide," and remember the sorting game or the article we read. This will help you remember what you've learned and done.

Optional Reflection Prompt (Teacher-Directed):

After completing the checklist, choose one of these questions and answer it in your notebook or share your thoughts aloud:

- "The thing I'm most proud of in this project is..."
- "One thing I want to get even better at, after this project, is..."
- "I learned that correctly sorting [Wet/Dry] Waste helps our environment by..."

Relation to Standards:

- **K-2-ETS1-2:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem.
    - (Students are designing a functional smart trash bin and lid using 3D shapes in Tinkercad, solving the problem of waste management.)
- **K-2-ETS1-3:** Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.
    - (Implicitly, students are designing solutions to the problem of waste segregation and collection, comparing different design choices for the bin, lid, or Super-Guide.)

1. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening):

- **SL.1.4:** Describe people, places, things, and events with relevant details, expressing ideas and feelings clearly.
    - (Students present and explain their smart bin designs and their "Super-Guides" for waste sorting.)
- **SL.1.5:** Add drawings or other visual displays to descriptions when appropriate to clarify ideas, thoughts, and feelings.
    - (The Tinkercad model of the smart bin and the "Super-Guide" serve as visual displays to clarify their design ideas and sorting instructions.)

2. Common Core State Standards (CCSS) - Mathematics (Geometry):

- **1.G.A.1:** Distinguish between defining attributes (e.g., number of sides, number of vertices) versus non-defining attributes (e.g., color, size, orientation); build and draw shapes to possess defining attributes.
    - (Students are using their understanding of 3D shapes and their attributes to construct components like the bin and its lid in Tinkercad.)